# Sarath Viswanathan M

*Senior Frontend Engineer | React · TypeScript · Three.js | Frontend Architecture | 6+ Years Experience*

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## PROFESSIONAL SUMMARY

**Senior Frontend Engineer with 6+ years of experience** building scalable, high-performance Single Page Applications (SPA) and enterprise web applications using **React.js, TypeScript, and modern JavaScript**. Strong expertise in **frontend architecture, state management, REST API integration, design systems, and performance optimization**. Experienced in building **Three.js-based interactive 3D/CAD visualization**, enterprise SaaS, and data-intensive applications. Led frontend development of an automated manufacturing costing platform that reduced quotation turnaround from **~3 days to under 5 minutes**. **Currently based in Abu Dhabi, UAE, and available to join immediately.**

## TECHNICAL SKILLS

**Frontend:** React.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, React Hooks, Redux Toolkit, React Context API, React Router, RemixJS, Formik, Yup, react-table

**UI & Visualisation:** Three.js, CAD/3D Visualisation, Chakra UI, Fluent UI, Tailwind CSS, Storybook

**Architecture & Performance:** Frontend Architecture, Design Systems, Component Architecture, Core Web Vitals, Code Splitting, Lazy Loading, Progressive Web Apps (PWA)

**AI-Assisted Development:** Cursor, Claude Code, ChatGPT Codex, AI-Assisted Code Analysis, Architecture Planning, Code Generation, Performance Optimisation, Debugging

**API & Real-Time Integration:** RESTful APIs, WebSockets, Axios, Frappe/ERPNext, Third-Party Integrations

**Microsoft Ecosystem:** Microsoft Teams Tab Apps, Teams JavaScript SDK (TeamsJS)

**Testing:** Cypress, Jest, React Testing Library

**Tooling & DevOps:** Vite, Webpack, Git, GitHub, CI/CD Pipelines

**Analytics & Observability:** Sentry, Datadog RUM, Google Analytics, Google Tag Manager

**Methodologies:** Agile, SAFe 5, Clean Coding Practices

## EXPERIENCE

**Karkhana.io** | Bengaluru *April 2025 – June 2026*

*Senior Full Stack Developer (Frontend Engineering) Promoted*

- **Owned frontend architecture** for Karkhana.io's Costing Engine, a large-scale React Single Page Application (SPA) built with TypeScript, Three.js, and Redux Toolkit, maintaining responsive interactive CAD visualisation as model complexity increased **10× in 12 months**.
- **Delivered 3–5 production features per quarter end-to-end**, covering frontend design, implementation, REST API integration, testing, and production release while maintaining **<2% post-release defect rate**.
- **Owned and scaled the company-wide React component library** using Chakra UI, Storybook, and Vite, expanding adoption across **4+ portals** and reducing cross-team UI development time by **~40%**.
- **Defined frontend performance standards** around Core Web Vitals, code splitting, and lazy loading, maintaining initial page load below **1.8 seconds** and Lighthouse scores above **90** as the product scaled.
- **Leveraged AI-assisted development tools** including Cursor, Claude Code, and ChatGPT Codex for codebase analysis, architecture planning, implementation, performance optimisation, and debugging, reviewing generated solutions for correctness and maintainability.
- **Represented Karkhana.io at IMTEX 2025**, delivering live product demonstrations, answering technical questions from industry users, and communicating complex product capabilities to non-technical audiences.

*Frontend Software Developer July 2022 – March 2025*

- **Architected and solely engineered the frontend** of Karkhana.io's Automated Costing Engine as an enterprise-scale Single Page Application (SPA) using React, TypeScript, Three.js, Redux, and Chakra UI, reducing customer quotation cycles from **~3 days to under 5 minutes**.
- **Built interactive Three.js-based CAD visualisation capabilities** for uploaded manufacturing models, including model rendering, user interaction, feature selection, and engineering-focused workflows within the browser.
- **Shipped 4+ production manufacturing portals** within 12 months using React, RemixJS, and TypeScript, contributing to **30% higher user engagement** and onboarding **200+ manufacturer accounts**.
- **Designed and standardised Redux Toolkit state architecture** across 4+ production portals, reducing state-related production defects by approximately **60%** and improving maintainability across frontend applications.
- **Integrated 60+ REST API endpoints** with a Frappe backend using TypeScript and Axios, reducing QA-reported integration defects by **~50%** and supporting weekly production releases.
- **Led Progressive Web App (PWA) enablement** across React manufacturing portals using service workers and caching strategies, improving reliability across mobile and low-bandwidth environments.

**GlobalLogic** | Noida *October 2019 – January 2022*

*Software Engineer (promoted from Trainee → Associate)*

- **Built 20+ reusable React components** for an enterprise CSR dashboard, standardising UI patterns across **3 product squads** and reducing feature development time by **~25%**.
- **Delivered 3+ React/Node.js prototypes and POCs**, reducing stakeholder feedback cycles from **2 weeks to under 4 days** and supporting **2 enterprise opportunities**.
- **Modernised legacy VB6/VB.NET modules through incremental React migration**, reducing recurring production issues by **~30%** without disrupting business operations.
- **Coordinated deliverables across a 12-person Agile squad**, improving on-time release performance from **~70% to 95%+ across 8 sprints**.

## EDUCATION

**Government Engineering College Wayanad, Kerala** *Aug 2015 – Jul 2019*

*B.Tech in Computer Science & Engineering | APJ Abdul Kalam Kerala Technological University*

## CERTIFICATION & RECOGNITION

- **Certified SAFe 5 Practitioner** — Scaled Agile Framework
- **Karkhana.io Best Performer of the Year** — Costing Engine delivery & component library rollout
- **GlobalLogic ACE of the Month** — Delivery leadership
- **GlobalLogic Spotlight of the Month** — Reusable frontend architecture
- **GlobalLogic ML Hackathon Championship** — First place

## LANGUAGES

English (Fluent) | Hindi (Professional Working Proficiency) | Malayalam (Native)